

CA One Report

Student ID: 20065898

Module: Programming (C# and Python)

Date: 08 April 2026

1. Overview

This assignment was completed in four parts as requested, with C# used for Section I (Que.1 and Que.2) and Python used for Section II (Que.3 and Que.4). The workspace was organized under the student-id folder (20065898) with separate subfolders for each question. The implementation focuses on object-oriented design, appropriate data structures, networking, persistence, and web scraping.

2. Que.1 (C#): Churros Food Truck Queue System

A menu-driven console application was implemented to simulate stall-owner operations for churros ordering and delivery.

Key features implemented:

- Displays churros menu with prices.
- Supports Place order, Deliver order, and Exit options.
- Generates incremental order numbers.
- Calculates payment before queue insertion.
- Uses FIFO order handling through Queue<Order>.

Object-oriented design:

- Churros class includes item name and price, with constructor and properties.
- Order class includes order number, order details, quantity, and bill.
- Order methods include placing an order, paying bill, and collecting order.
- Encapsulation is maintained with private state (e.g., payment status) and controlled property access.

Testing:

- Unit test project (ChurrosApp.Tests) verifies PayBill() returns correct total and updates Bill property.
- Test result: passed.

3. Que.2 (C#): Periodic Table Elements (1-30)

A C# console program was created to store and retrieve the first 30 periodic table elements by atomic number.

Implementation details:

- Data structure used: Dictionary<int, Element>.
- Element record stores atomic number, element name, class, and short information.
- User provides atomic number input.
- Program validates input and displays details when present.
- Loop continues with y and terminates on n.

This satisfies the requirement to provide element information via atomic number using a suitable structure.

4. Que.3 (Python): EasyDrive TCP Client-Server Registration

A connection-oriented (TCP) client-server system was implemented.

Client responsibilities:

- Collect customer name, address, PPS number, and driving license data.
- License can be submitted as file content (base64) or text input.
- Sends registration request as JSON to server.

Server responsibilities:

- Accepts TCP connections.
- Validates request payload.
- Stores registration in SQLite (disk-persistent relational database).

- Generates and returns unique registration number (ED-YYYYMMDD-#####).

Persistence:

- Database file: Que.3/easydrive.db.
- Table stores customer data, license data, timestamps, and registration numbers.

Verification:

- Successful connection and registration response were tested.
- Database query confirmed persisted records.

File naming note:

- Existing files Que4_server.py and Que4_client.py were retained.
- Added Que3_server.py and Que3_client.py entrypoints for clearer question mapping.

5. Que.4 (Python): Travel Books Web Scraper

A Python scraper was developed for:

https://books.toscrape.com/catalogue/category/books/travel_2/index.html

Implementation:

- Uses requests to fetch HTML.
- Uses BeautifulSoup to parse book entries.
- Extracts book title, rating, and price.
- Saves output to CSV (travel_books.csv).
- Reads all rows from CSV and prints to terminal at program end.

Verification:

- Dependencies installed (requests, beautifulsoup4).
- Script executed successfully and printed all rows from CSV.

6. Folder Organization and Submission Notes

The work is organized into separate folders (Que.1, Que.2, Que.3, Que.4) under the student-id main folder. This aligns with structural requirements.

Before final Moodle submission, the following must be attached manually:

- Electronic Assignment Cover Sheet.
- GitHub repository link: <https://github.com/syedhadihussain5125/20065898>

7. External Help and Resources Attributed

- 1. Microsoft Learn - .NET, C#, and MSTest documentation.**
- 2. Python official documentation (socket, sqlite3, csv, json, base64).**
- 3. Requests documentation for HTTP retrieval.**
- 4. Beautiful Soup documentation for HTML parsing.**
- 5. Wikipedia periodic table reference for element data context.**
- 6. PubChem element pages for concise element information.**

8. Screenshot Evidence

Que.1 output screenshot:

```
PS C:\Users\syedh\Documents\20065898\Que.1> dotnet -ChorusApp\bin\Debug\net8.0\ChorusApp.dll
-----
Delicious Churros:
1. Churros with plain sugar: EUR 6.00
2. Churros with cinnamon sugar: EUR 6.00
3. Churros with chocolate sauce: EUR 8.00
4. Churros with Nutella: EUR 8.00
-----
1. Place order
2. Deliver order
0. Exit
-----
Choose your option: 1
Select churros type (1-4): 2
Enter quantity: 3
Order #1 placed: 3 x Churros with cinnamon sugar
Success: your order has been placed.
Payment successful. Total bill: EUR 18.00
Order #1 added to queue.
Orders waiting: 1

-----
Delicious Churros:
1. Churros with plain sugar: EUR 6.00
2. Churros with cinnamon sugar: EUR 6.00
3. Churros with chocolate sauce: EUR 8.00
4. Churros with Nutella: EUR 8.00
-----
1. Place order
2. Deliver order
0. Exit
-----
Choose your option: 2
Order #1 collected successfully.
Delivered item: Churros with cinnamon sugar x 3
Remaining orders: 0

-----
Delicious Churros:
1. Churros with plain sugar: EUR 6.00
2. Churros with cinnamon sugar: EUR 6.00
3. Churros with chocolate sauce: EUR 8.00
4. Churros with Nutella: EUR 8.00
-----
1. Place order
2. Deliver order
0. Exit
-----
Choose your option: 0
System closed. Goodbye.
PS C:\Users\syedh\Documents\20065898\Que.1>
```

Que.2 output screenshot:

```
PS C:\Users\syedh\Documents\20065898\Que.2> dotnet run
Hi there! Happy to help!
Provide atomic number of the element: 12
Atomic Number: 12
Name: Magnesium
Class: Alkaline Earth Metal (Magnesium is important in alloys, medicine, and chlorophyll.)
Do you want to know more elements [y/n]? y
Provide atomic number of the element: 10
Atomic Number: 10
Name: Neon
Class: Noble Gas (Neon emits a bright glow in electric discharge tubes and signage.)
Do you want to know more elements [y/n]? y
Provide atomic number of the element: 30
Atomic Number: 30
Name: Zinc
Class: Transition Metal (Zinc is used for galvanizing steel and in biological enzyme functions.)
Do you want to know more elements [y/n]?
```

Que.3 client-server and database screenshots:

```
PS C:\Users\syedh\Documents\Que.3> & ".\env\Scripts\python.exe"
".\Que4_client.py"
=== EasyDrive Customer Registration ===
Enter Name: Syed Hadi
Enter Address: 45 Riverwalk Apartments, Dublin 4, Ireland
Enter PPS Number: 1234567TA

Driving License Document:
- Enter a file path to upload the document
- Or press Enter to type document details manually
License file path (optional):
Enter driving license details: MURPH12345678

Connecting to server 127.0.0.1:5000...
Connection established.

Registration Successful
Your unique registration number is:
ED-20260325-000002
PS C:\Users\syedh\Documents\Que.3>
```

```

PS C:\Users\syedh\Documents\Que.3> & ".\venv\Scripts\python.exe" ".\Que4_server.py"
EasyDrive server is listening on 127.0.0.1:5000
Database file: C:\Users\syedh\Documents\Que.3\easydrive.db
[+] Connected by ('127.0.0.1', 51218)
[+] Registration completed for ('127.0.0.1', 51218). Reg No: ED-20260325-000002
[-] Connection closed for ('127.0.0.1', 51218)

```

Database Structure Browse Data Edit Pragma Execute SQL						
Table: customers						
	id	registration_number	name	address	pps_number	license_filename
1	2	ED-20260325-000002	Syed Hadi	45 Riverwalk Apartments, Dublin 4, Ireland	1234567TA	NULL
2	1	ED-20260325-000001	Alice Murphy	12 Main Street, Dublin	PPS12345A	NULL

Database Structure Browse Data Edit Pragma Execute SQL						
Table: customers						
	address	pps_number	license_filename	license_blob	license_text	created_at
1	45 Riverwalk Apartments, Dublin 4, Ireland	1234567TA	NULL	NULL	MURPH12345678	2026-03-25T17:48:09
2	12 Main Street, Dublin	PPS12345A	NULL	NULL	DL-IRL-998877	2026-03-25T17:29:01

DB Browser for SQLite - C:\Users\syedh\Documents\20065898\Que.3\easydrive.db		
File Edit View Tools Help		
Database Structure Browse Data Edit Pragma Execute SQL		
Table: sqlite_sequence		
	name	seq
1	customers	2

Que.4 scraper results screenshot:

D6			
	A	B	C
1	book_name	rating	price
2	It's Only the Himalayas	2	£45.17
3	Full Moon over Noah's Ark: An Odyssey to Mount Ararat and Beyond	4	£49.43
4	See America: A Celebration of Our National Parks & Treasured Sites	3	£48.87
5	Vagabonding: An Uncommon Guide to the Art of Long-Term World Travel	2	£36.94
6	Under the Tuscan Sun	3	£37.33
7	A Summer In Europe	2	£44.34
8	The Great Railway Bazaar	1	£30.54
9	A Year in Provence (Provence #1)	4	£56.88
10	The Road to Little Dribbling: Adventures of an American in Britain (Notes From a Small Island #2)	1	£23.21
11	Neither Here nor There: Travels in Europe	3	£38.95
12	1,000 Places to See Before You Die	5	£26.08
13			
14			
15			
16			